Predictive Modeling of Mental Wellbeing in Finnish Immigrants

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Project Overview & Objectives

Goal

- Predict mental well-being indicators from passive sensor and wearable data
- Dataset: 28 days of multimodal data from 39 Finnish immigrant participants
- Source: "Loneliness and Well-being in Finnish Immigrants: A Multimodal Dataset from Wearables and Passive Data Collection"

Learning Objectives

- Explore and preprocess a real-world multimodal health dataset
- Engineer meaningful daily features from raw sensor logs
- Train and compare SVM, Random Forest, XGBoost, and Neural Network
- Evaluate model performance and interpret results

Dataset Overview — 4 Multimodal Sources

Smartphone Behavioral Data (AWARE App)

- Screen usage, notifications, calls, messages, battery

Wearable Device Data (Oura Ring)

- Sleep duration, sleep stages, sleep quality, steps, activity, heart rate variability (HRV)

Smartwatch Sensor Data (Samsung Watch)

- Heart rate, PPG (blood volume data), accelerometer, gyroscope, pressure

Survey & Psychological Data

- PHQ-9 (depression), PSS (stress), UCLA Loneliness Scale, Social Connectedness, EMA daily mood

Structure

- 39 participants * up to 28 days
- Folders: Aware, Oura, Watch, Surveys

Task 1: Paper Review

Q1 — Purpose

- Study loneliness and well-being in first-generation Finnish immigrants via passive and wearable sensing
- Enable research linking behavioral/physiological signals to mental health outcomes

Q2 — Data Types

- Smartphone behavior (AWARE)
- Wearable physiology (Oura Ring)
- Raw sensor signals (Samsung Watch)
- Psychological surveys

Q3 — Research Applications

- Predicting depression, stress, and loneliness from passively collected data
- Mental health monitoring for at-risk and immigrant populations
- Digital phenotyping and multimodal sensor fusion research

Task 2: Data Exploration

- 39 participants, IDs non-consecutive (1–46 with gaps)
- 2,083 EMA observations with valid loneliness scores across all participants
- EMA collected multiple times per day (up to 4–5x)

Known gaps

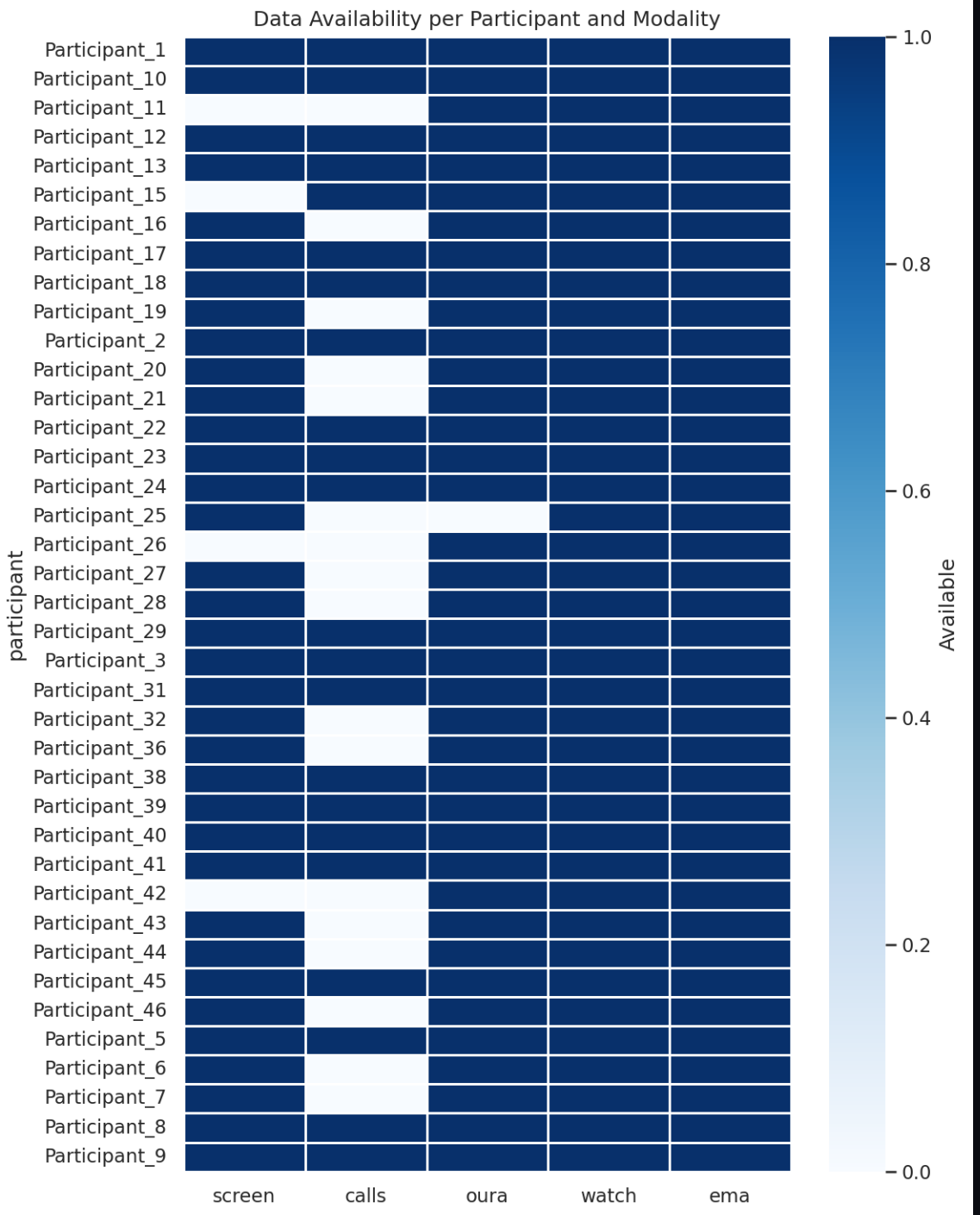
- P11, P26, P42 have no AWARE data
- P25 has no Oura data

Missing rates

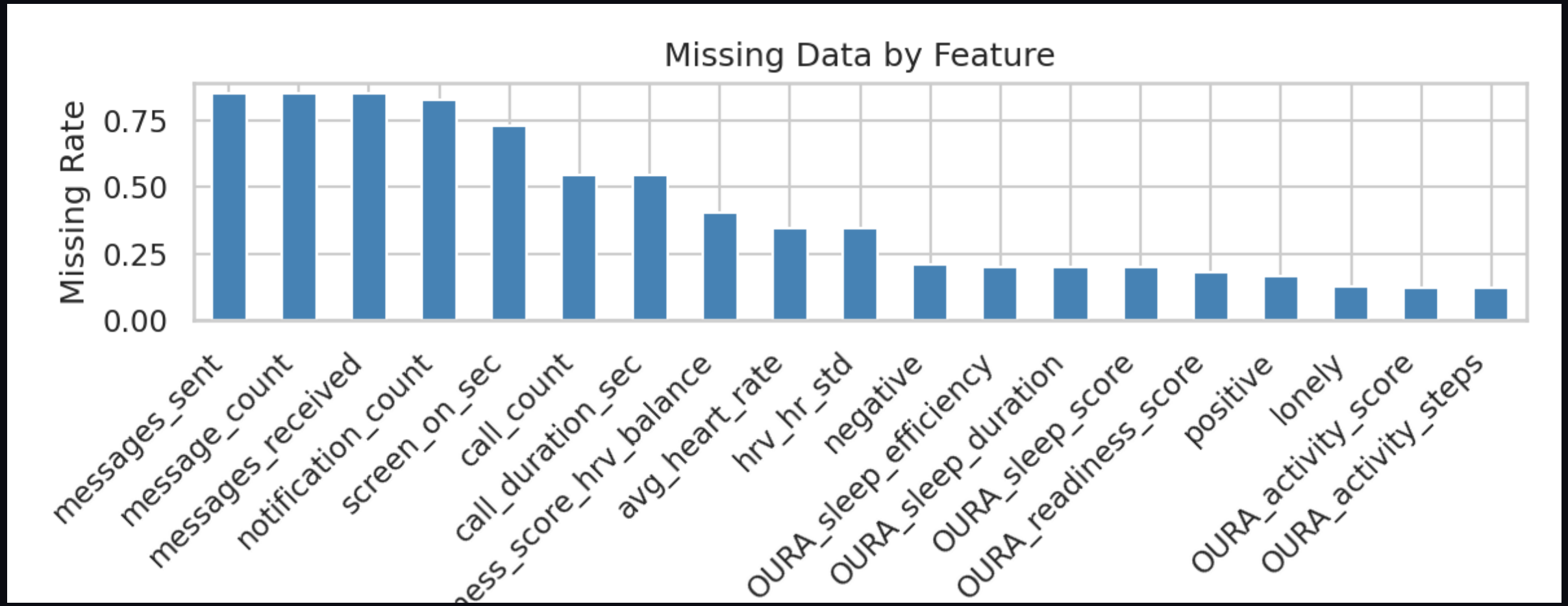
- Messages/notifications ~84%
- Screen ~73%
- Calls ~55%
- Oura sleep ~20%
- Watch HR ~35%

4/8/2026

Data Availability



Missing Data



Task 3.1: Data Preprocessing

Timestamp conversion

- 13-digit Unix ms → calendar date using pd.to_datetime(..., unit='ms')

Watch cleaning

- Removed HRM values ≤ 0 (sensor noise artifacts)

Record filtering

- Dropped rows with missing EMA loneliness target

Missing feature imputation

- Column-wise median via SimpleImputer

Normalization

- StandardScaler (zero mean, unit variance) applied before SVM and MLP

Task 3.2: Feature Engineering

Final feature matrix

- 16 daily features across all 4 modalities

Behavioral (AWARE)

- Screen-on time
- Call count, call duration
- Message count, messages sent, messages received
- Notification count

Sleep & Activity (Oura)

- Sleep duration, sleep efficiency, sleep score
- Daily steps, activity score
- Readiness score, HRV balance

Physiological (Samsung Watch)

- Daily average heart rate, heart rate variability (HR std)
- Aggregated across 6,858 CSV files total

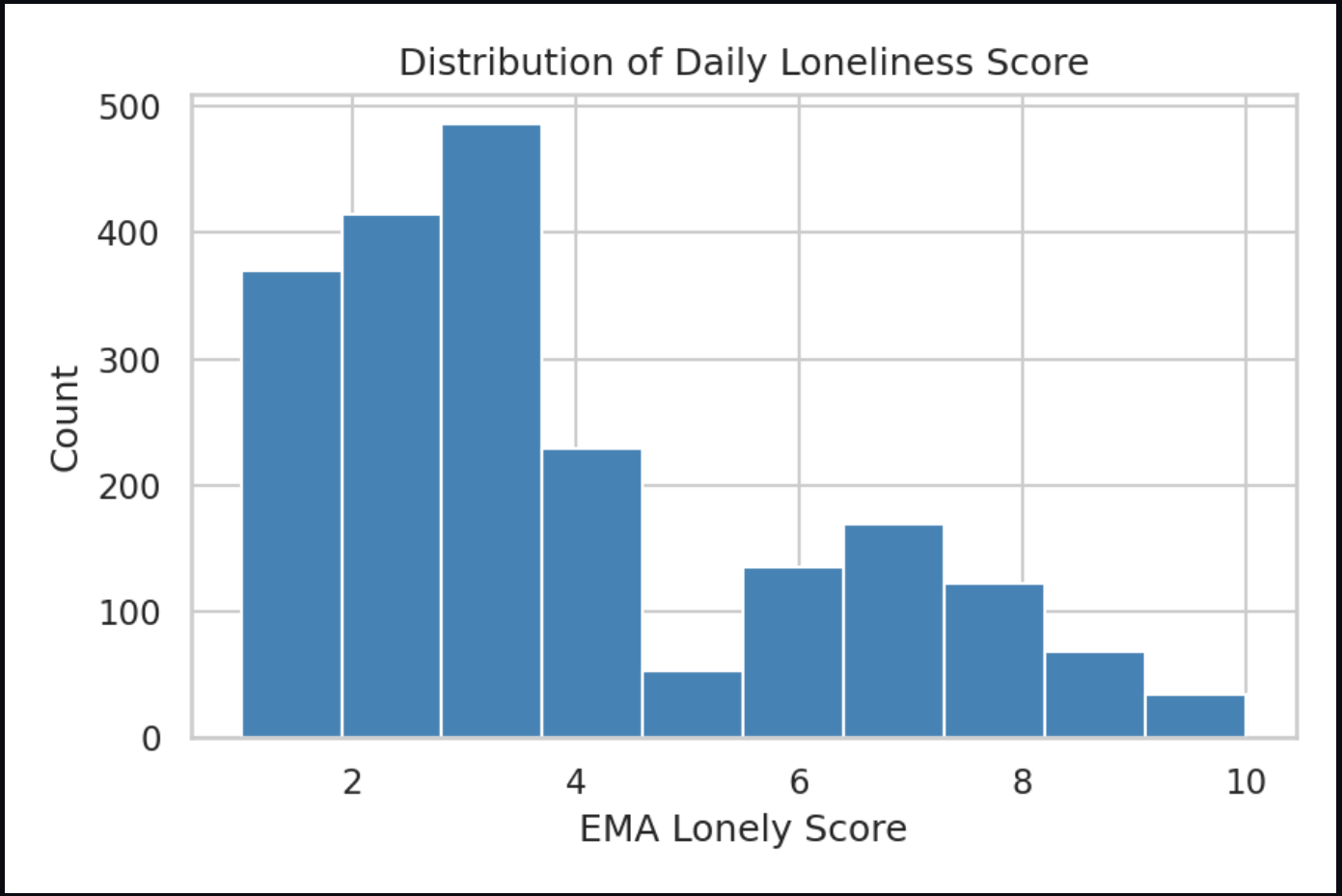
Merged all features per participant per day

- One row per EMA response

Task 4: Prediction Task Definition

- Supervised binary classification problem
- Target
  - EMA daily loneliness score
  - 1–10 scale, self-reported
- Binarized at median score of 3.0
- High loneliness (score > 3)
  - 812 observations
- Low loneliness (score ≤ 3)
  - 1,271 observations
- Train/test
  - 80/20 stratified split (random_state = 42)

Loneliness Score Distribution



Task 5: ML Models — SVM & Random Forest

Support Vector Machine (SVM)

- Finds the maximum-margin hyperplane separating high/low loneliness classes
- Kernel: RBF (non-linear separation); tune C and gamma via GridSearchCV
- Requires feature scaling — apply StandardScaler before fitting
- *sklearn.svm.SVC(kernel = 'rbf', probability = True)*

Random Forest

- Ensemble of decision trees trained on random feature/sample subsets (bagging)
- Robust to outliers; provides built-in feature importances
- Tune: n_estimators, max_depth, min_samples_split via cross-validation
- *sklearn.ensemble.RandomForestClassifier(n_estimators = 200)*

Task 5: ML Models — XGBoost & Neural Network

XGBoost

- Gradient boosting: sequentially trains trees to correct prior residuals
- Often top-performing on tabular data; handles class imbalance via scale_pos_weight
- Tune: n_estimators, learning_rate, max_depth, subsample
- *xgboost.XGBClassifier(eval_metric = 'logloss')*

Neural Network (MLP)

- Multi-layer Perceptron with ReLU activations and sigmoid output for binary classification
- Architecture: Input → Dense(64, ReLU) → Dropout(0.3) → Dense(32, ReLU) → Dense(1, Sigmoid)
- Train with Adam optimizer, binary cross-entropy loss, early stopping on val_loss
- *sklearn.neural_network.MLPClassifier or TensorFlow/Keras Sequential model*

Task 6: Model Evaluation

Evaluation Strategy

- Train/Test Split: 80/20 stratified split to preserve class balance
- Cross-Validation: 5-fold StratifiedKFold for robust performance estimates (small dataset)

Metrics Reported for Each Model

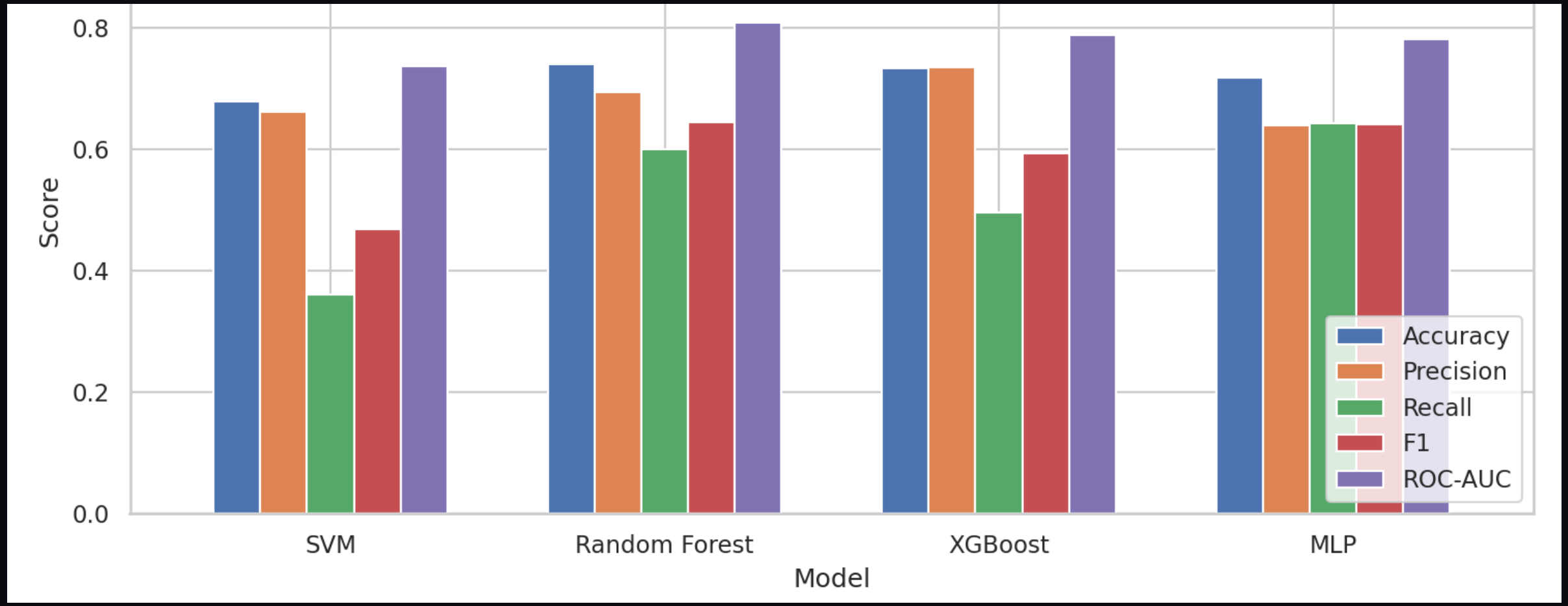
- **Accuracy:** Overall fraction of correct predictions
- **Precision & Recall:** Especially important given potential class imbalance
- **F1 Score:** Harmonic mean of precision and recall — primary metric
- **ROC-AUC:** Area under ROC curve; measures discrimination regardless of threshold

Deliverable: Comparison table of all 4 models across all 5 metrics + ROC curve plot

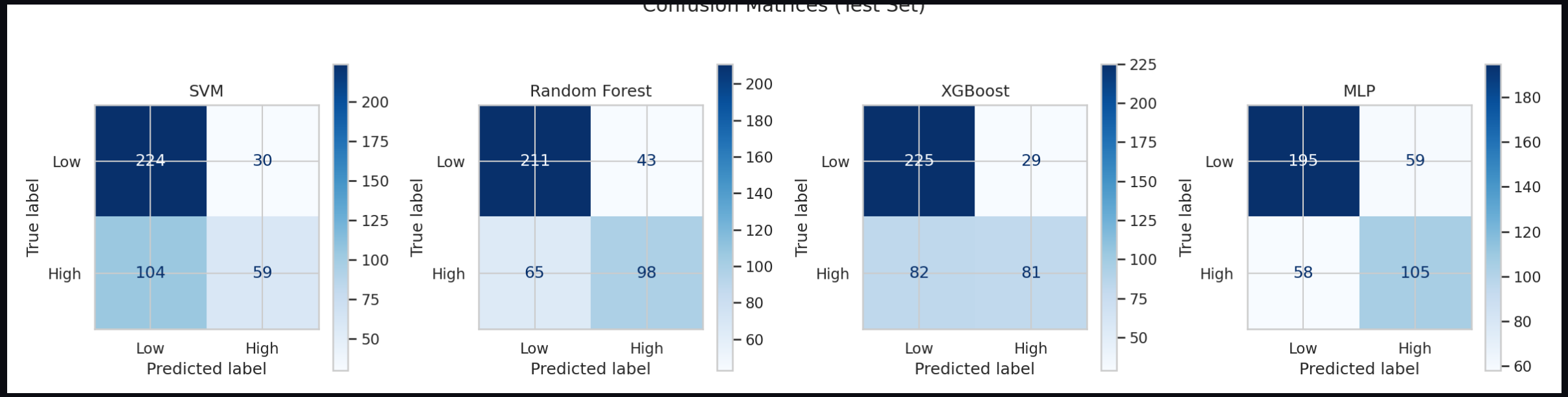
Evaluation Results

Model	Accuracy	Precision	Recall	F1	ROC-AUC	CV F1 (mean ± std)
Random Forest	0.741	0.695	0.601	0.645	0.809	0.677 ± 0.009
XGBoost	0.734	0.736	0.497	0.593	0.788	0.624 ± 0.011
MLP	0.719	0.640	0.644	0.642	0.782	0.666 ± 0.016
SVM	0.679	0.663	0.362	0.468	0.737	0.528 ± 0.028

Model Comparison



Confusion Matrix



Task 7: Result Analysis & Key Insights

Best model

- Random Forest (F1 = 0.645, ROC-AUC = 0.809, CV F1 = 0.677)

Top 5 Features

- Daily steps
- Avg heart rate
- Sleep duration
- Activity score
- HRV std

Physical/physiological features outperformed digital behavioral features

SVM weakest — recall of 0.362, biased toward majority (low loneliness) class

Passive wearable sensing alone can meaningfully predict momentary loneliness

Limitations: n = 39, EMA pseudo-replication, no participant-stratified split